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MIL-E-1/746G
24 April 1973
SUPERSEDING
MIL-E-1/746F
31 August 1966
(See note 12)

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

1/ TYPE 6BM6A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Reflex, separate cavity, pulse

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Ec3	Er	Ik	Ehk	Pi	tk	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	mAdc	Vdc	W	sec	ft
Maximum:	6.9	+1.0	350	350	-700	29	+45	10	---	10,000
Minimum:	5.7	-500	---	---	-15	---	-45	---	120	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1
Cathode:	Coated unipotential
Mounting position:	Any

TEST CONDITIONS:

Parameter:	Ef	Ec1	Ec2	Ec3	Er
Unit:	V	Vdc	Vdc	Vdc	Vdc
Test condition 1:	6.3	0	325	325	Adj for max Po
Test condition 2:	6.3	0	300	300	Adj for max Po

GENERAL:

Qualification - Required

1/ CAUTION: Tube not designed for applications where hysteresis may affect performance.

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	1	-					
1256	Electrode current (cathode)	-	-	Ec2 = Ec3 = 325 Vdc; Ec1 = 0; Er = 0	Ik	15	24	mAdc
4229	Total reflector current (1)	4	-	Er = -90 Vdc	Ir	---	5	µAdc
1301	Heater current	-	-		If	500	750	mA
1336	Heater-cathode leakage	-	-	Ehk = ±45 Vdc	Ihk	0	75	µAdc
4213	Reflector voltage (1)	5	1		Er	-30	-70	Vdc
---	Frequency (1)	5	1		F	795	825	MHz
4250	Power output (1)	5	1		Po	50	---	mW
4213	Reflector voltage (2)	6	1		Er	-450	-550	Vdc
---	Frequency (2)	6	1		F	2,150	2,250	MHz
4250	Power output (2)	6	1		Po	50	---	mW
4214	Cathode emission	7	1		ΔIk/Ik	---	15	%
4229	Total reflector current (2)	5	1		Ir	---	50	µAdc
4213	Reflector voltage (3)	8	2		Er	-20	-60	Vdc
---	Frequency (3)	8	2		F	1,130	1,170	MHz
4250	Power output (3)	8	2		Po	10	---	mW
4229	Total reflector current (3)	8	2		Ir	---	50	µAdc
4213	Reflector voltage (4)	9	2		Er	-275	-355	Vdc
---	Frequency (4)	9	2		F	2,680	2,820	MHz
4250	Power output (4)	9	2		Po	20	---	mW
1256	Electrode current (cathode cutoff)	-	-	Ec2 = Ec3 = 325 Vdc; Er = 0; Ec1 = -150 Vdc	Ik	---	5	mAdc
	Quality conformance inspection, part 2							
1101	Secureness of base, cap, or insert	10	-		---	---	---	---
1031	Low-frequency vibration	-	-	No voltages	---	---	---	---
4318	Time jitter	3, 11	-	Prr = 4,000 (approx)	tj	---	0.2	µs

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3							
---	Life test	2	1	Group C	t	500	---	hrs
---	Life-test end points:							
4250	Power output (3)	8	2		Po	8	---	mW
4250	Power output (4)	9	2		Po	16	---	mW

NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.
2. Test tube with 1,000 MHz test cavity as specified on Drawing 166-JAN. Frequency approximately 1,000 MHz. Operate tube in N = 1 mode (1-3/4 cycles for complete transit). Adjust Er for maximum power output. Adjust output coupling loop for maximum power output.
3. Test in signal generator TS-419/U, or equivalent.
4. Static test; tube not oscillating.
5. Test tube with 800 MHz test cavity as specified on Drawing 166-JAN. Adjust cavity to resonate at 800.5 MHz with standard plug as specified on Drawing 197-JAN. Operate tube in N = 1 mode (1-3/4 cycles for complete transit). Adjust Er for maximum power output. Adjust output coupling loop for maximum power output.
6. Test tube with 2,200 MHz test cavity as specified on Drawing 166-Jan. Adjust cavity to resonate at 2,113.8 MHz with standard plug as specified on Drawing 197-JAN. Operate tube in N = 1 mode (1-3/4 cycles for complete transit). Adjust Er for maximum power output. Adjust output coupling loop for maximum power output.
7. Reduce Ef to 5.7 V while tube is oscillating. The cathode emission shall not exceed the limit shown.
8. Test tube with 1,150 MHz test cavity as specified on Drawing 166-JAN. Adjust cavity to resonate at 1,130.6 MHz with standard plug as specified on Drawing 197-JAN. Operate tube in N = 2 mode (2-3/4 cycles for complete transit). Adjust Er for maximum power output. Adjust output coupling loop for maximum power output.
9. Test tube with 2,750 MHz test cavity as specified on Drawing 166-JAN. Adjust cavity to resonate at 2,569.2 MHz with standard plug as specified on Drawing 197-JAN. Operate tube in N = 2 mode (2-3/4 cycles for complete transit). Adjust Er for maximum power output. Adjust output coupling loop for maximum power output.
10. Water-immersion test omitted. Torque shall be applied between upper disk (g3) and molded portion of base and shall be 20 inch-pounds.
11. Adjust Er tracking for minimum jitter over the range from 900 to 2,100 MHz. When recheck measurements are made at locations other than the point of manufacture, an addition of 0.05 µs shall be allowed on the specified limits.
12. Changes from previous issue. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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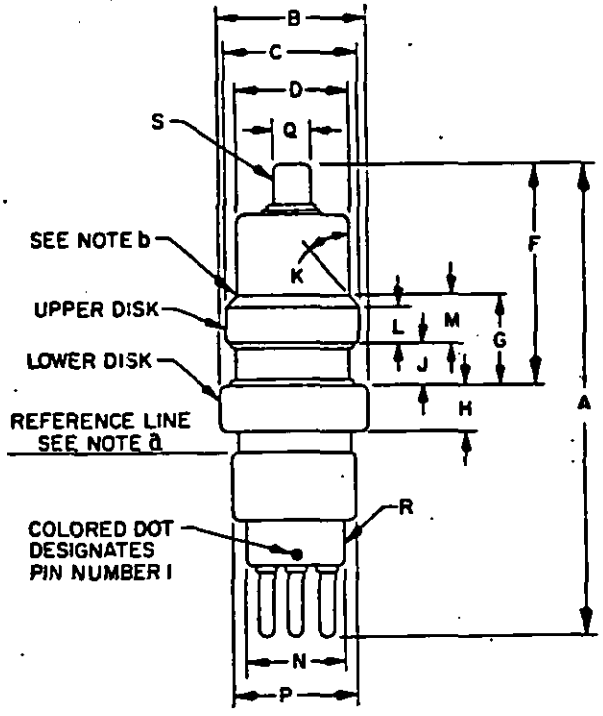
Preparing activity: Navy - EC

Agent: DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A	2.969 (75.41)	3.219 (81.76)
B	.990 (25.15)	1.000 (25.40)
C	.880 (22.35)	.890 (22.61)
F	1.375 (34.93)	1.500 (38.10)
M	.303 (7.70)	.323 (8.20)
Quality conformance inspection, part 2		
G	.560 (14.22)	.610 (15.49)
H	.250 (6.35)	.312 (7.92)
J	.257 (6.53)	.287 (7.29)
Quality conformance inspection, part 3		
D		.750 (19.05)
K	25°	35°
N	.610 (15.49)	.656 (16.66)
P	.807 (20.50)	.822 (20.88)
Q	.245 (6.22)	.255 (6.48)
R	Base: A4-76	
S	Cap: C1-3	
Nominal dimensions		
L	.250 (6.35)	



PIN CONNECTIONS

1	g1
2	h
3	k
4	h
Lower disk	g2
Upper disk	g3
Cap	r

NOTES:
a. For dimensions below reference line refer to base A4-76.
b. Space between glass and upper disk edge or backing ring shall be .030 maximum. Quality conformance inspection, part 2 shall apply.
c. Diameters B and C shall be concentric within .025 TIR and shall form circular cylinder within a range of .006. Allowable radius at ends of cylinder shall be .030 maximum. Quality conformance inspection, part 2, shall apply.
d. Tube shall be tested in concentricity gage specified on drawing 199-JAN. Gage readings on diameters shall be as follows:

Diameter	Eccentricity
D	.050 max
N	.060 max
P	.060 max
Q	.040 max

e. All concentricity measurements shall be quality conformance inspection, part 2.
f. Pin No. 1 shall be identified by an index mark (colored dot or other).

FIGURE 1. Outline drawing of electron tube type 6BM6A.

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NAVAL ELECTRONIC SYSTEMS COMMAND
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